

PRISM - P FILE

Problem Statement

In many hospitals and clinics, appointment scheduling is still done manually or using basic systems. Doctors' availability is not properly managed, and patients often face delays.

- Leads to long waiting times
- Causes double bookings or missed appointments
- Inefficient use of doctor time
- Poor patient experience
- **Patients:** Long waits, inconvenience
- **Doctors:** Unbalanced workload
- **Hospital Staff:** Manual effort, errors

Business Objective

The primary objective is to develop an **AI-powered healthcare scheduling system** that:

- Automates appointment booking
- Reduces scheduling conflicts
- Improves doctor time utilization
- Enhances patient experience

Success Metrics

Leading Indicators

- Number of appointments booked
- Less manual work by staff
- More slots getting filled
- Slot utilization rate

Lagging Indicators

- Waiting time reduces
- No double bookings
- Patients are more satisfied

Stakeholders & Decision Owner

Type	Name	Responsibility
Stakeholder	Hospital Staff	Manage patient records
Stakeholder	Doctors	Set availability & consultation slots
Stakeholder	Patients	Book appointments
Stakeholder	Developers	Develop & maintain system
Decision Owner	Hospital Administration / Project Lead	Final decisions & system oversight

Doctors Table

doctor_id	name	available_from	available_to	slot_duration
101	Dr. Sharma	10:00	12:00	30 mins

Patients Table

patient_id	name	age	contact
1	Ravi	25	9876543210
2	Anjali	30	9123456780

Appointments Table

appointment_id	patient_id	doctor_id	time_slot	status
1	1 (Ravi)	101	10:30	Booked

Current State (As-Is)

- Patient details are recorded manually
 - Staff writes or enters data in basic systems, which increases chances of errors and duplication.
- Doctors inform availability informally
 - Availability is shared verbally or through messages, leading to lack of proper tracking.
- Appointments are scheduled manually or via phone
 - Staff handles bookings manually, which is time-consuming and inefficient.
- No structured slot system (30 min / 1 hour)
 - Appointments are not divided into fixed time slots, causing poor time management.

- High chances of overlap and inefficiency
 - Multiple bookings at the same time and uneven workload for doctors.

Pain Points

Fragmented patient data across systems

→ Patient information is stored in multiple places (paper, spreadsheets, different tools), making it difficult to access complete records quickly.

Inefficient manual scheduling process

→ Staff-dependent booking leads to delays, errors, and increased operational workload.

Frequent appointment conflicts & double bookings

→ Lack of structured slot management causes overlapping appointments and doctor overbooking.

Underutilization or overloading of doctors

→ Some doctors are overbooked while others remain underutilized due to poor scheduling visibility.

Lack of real-time availability tracking

→ Changes in doctor schedules are not updated instantly, leading to outdated booking information.

Poor patient experience and long wait times

→ Patients face uncertainty, long queues, and inefficient service flow.

No automated reminders or follow-ups

→ Results in missed appointments and revenue loss for hospitals.

Limited scalability of current systems

→ Manual or semi-digital systems cannot handle increasing patient volume efficiently.

Lack of data-driven decision making

→ No analytics to optimize schedules, predict demand, or improve operations.

High dependency on human intervention

→ Entire workflow depends heavily on staff, increasing chances of errors and inefficiencies.

Constraints & Non-Negotiables

- **Time Constraint**

- The system must be developed within a limited timeframe (e.g., hackathon or project deadline), so only core features will be prioritized.

- **Budget Constraint**

- The solution should use low-cost or free technologies (open-source tools) to minimize development and deployment costs.

- **Compliance (Data Privacy & Security)**

- Patient data must be handled securely, ensuring confidentiality and basic healthcare data protection standards.

- **Technology Constraint**

- The system will be limited to a web-based stack (React, Flask/Spring Boot, MySQL), avoiding complex or heavy infrastructure.

- **Integration Limitations**

- Initial version will not integrate with external hospital systems (like EMR/EHR), and will function as a standalone application.